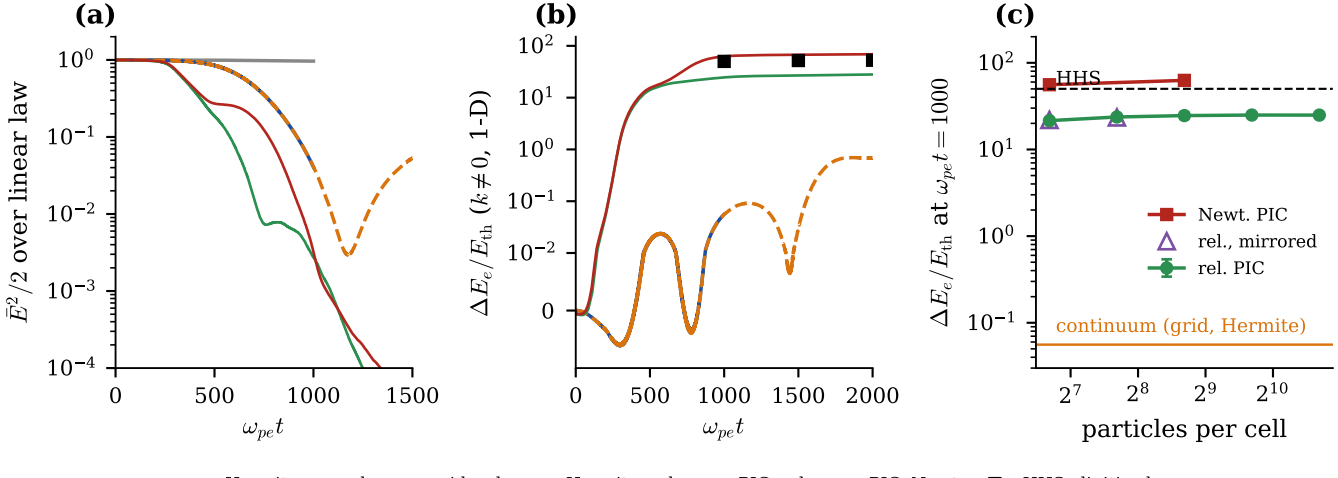

(a)
$\bar{E}^2/2$ over linear law
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
500
1000
1500
$\omega_{pe}t$
(b)
$\Delta E_e/E_{\rm th}$ ($k \neq 0$, 1-D)
10^2
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
$\omega_{pe}t$
(c)
$\Delta E_e/E_{\rm th}$ at $\omega_{pe}t = 1000$
10^2
10^1
10^0
10^{-1}
HHS
Newt. PIC
rel., mirrored
rel. PIC
continuum (grid, Hermite)
2^7
2^8
2^9
2^{10}
particles per cell
Hermite, nonrel.
grid, rel.
Hermite, rel.
PIC, rel.
PIC, Newt.
HHS, digitized